

E-COMMERCE SQL PROJECT

Sample Dataset & Project Reference

This dataset is designed for a MySQL/SQL analytics project. It contains customer, product, order, payment, discount and delivery information that can be used for JOINS, GROUP BY, aggregate functions, subqueries, CTEs and window functions.

Dataset size: 120 orders | 40 customers | 20 products

Period: January–December 2025

1. Customers Table

customer_id	customer_name	email	city	state
1	Arjun Sharma	customer001@example.com	Gurugram	Haryana
2	Yash Singh	customer002@example.com	Bengaluru	Karnataka
3	Rohan Verma	customer003@example.com	Noida	Uttar Pradesh
4	Kavya Verma	customer004@example.com	Ahmedabad	Gujarat
5	Neha Sharma	customer005@example.com	Lucknow	Uttar Pradesh
6	Aditya Singh	customer006@example.com	Bengaluru	Karnataka
7	Pooja Srivastava	customer007@example.com	Lucknow	Uttar Pradesh
8	Kavya Singh	customer008@example.com	Gurugram	Haryana
9	Kavya Jain	customer009@example.com	Bengaluru	Karnataka
10	Riya Srivastava	customer010@example.com	Hyderabad	Telangana
11	Aarav Gupta	customer011@example.com	Gurugram	Haryana
12	Neha Mishra	customer012@example.com	Hyderabad	Telangana
13	Rohan Singh	customer013@example.com	Pune	Maharashtra
14	Arjun Verma	customer014@example.com	Jaipur	Rajasthan
15	Arjun Mishra	customer015@example.com	Pune	Maharashtra
16	Simran Khan	customer016@example.com	Lucknow	Uttar Pradesh
17	Riya Patel	customer017@example.com	Delhi	Delhi
18	Priya Verma	customer018@example.com	Chennai	Tamil Nadu
19	Ananya Srivastava	customer019@example.com	Pune	Maharashtra
20	Meera Singh	customer020@example.com	Gurugram	Haryana
21	Aditya Sharma	customer021@example.com	Noida	Uttar Pradesh
22	Kunal Khan	customer022@example.com	Delhi	Delhi
23	Kunal Verma	customer023@example.com	Jaipur	Rajasthan
24	Yash Agarwal	customer024@example.com	Noida	Uttar Pradesh
25	Ishita Gupta	customer025@example.com	Pune	Maharashtra
26	Ishita Singh	customer026@example.com	Noida	Uttar Pradesh

customer_id	customer_name	email	city	state
27	Yash Verma	customer027@example.com	Ahmedabad	Gujarat
28	Kabir Patel	customer028@example.com	Gurugram	Haryana
29	Kunal Gupta	customer029@example.com	Kolkata	West Bengal
30	Priya Khan	customer030@example.com	Noida	Uttar Pradesh
31	Kavya Singh	customer031@example.com	Noida	Uttar Pradesh
32	Aanya Sharma	customer032@example.com	Bengaluru	Karnataka
33	Vivaan Mishra	customer033@example.com	Jaipur	Rajasthan
34	Yash Verma	customer034@example.com	Bengaluru	Karnataka
35	Meera Mishra	customer035@example.com	Bengaluru	Karnataka
36	Sneha Jain	customer036@example.com	Noida	Uttar Pradesh
37	Riya Gupta	customer037@example.com	Hyderabad	Telangana
38	Rohan Singh	customer038@example.com	Gurugram	Haryana
39	Kavya Patel	customer039@example.com	Hyderabad	Telangana
40	Meera Jain	customer040@example.com	Ahmedabad	Gujarat

2. Products Table

product_id	product_name	category	unit_price
1	Wireless Headphones	Electronics	■2,499.00
2	Bluetooth Speaker	Electronics	■1,799.00
3	Smart Watch	Electronics	■3,299.00
4	USB-C Cable	Electronics	■499.00
5	Power Bank	Electronics	■1,299.00
6	Laptop Backpack	Accessories	■1,499.00
7	Running Shoes	Fashion	■2,299.00
8	Cotton T-Shirt	Fashion	■699.00
9	Denim Jeans	Fashion	■1,599.00
10	Formal Shirt	Fashion	■1,199.00
11	Coffee Maker	Home & Kitchen	■3,499.00
12	Air Fryer	Home & Kitchen	■5,299.00
13	Water Bottle	Home & Kitchen	■799.00
14	Non-Stick Pan	Home & Kitchen	■1,399.00
15	Yoga Mat	Sports	■999.00
16	Dumbbells Set	Sports	■1,899.00
17	Cricket Bat	Sports	■2,499.00
18	Face Wash	Beauty	■449.00
19	Sunscreen SPF 50	Beauty	■699.00
20	Shampoo	Beauty	■549.00

3. Orders Dataset

The order-level data below is also provided separately as a UTF-8 CSV for direct MySQL import.

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Ishita Singh	Noida	Cotton T-Shirt	Fashion	1	699	15	594.15	UPI
Kavya Singh	Gurugram	Laptop Backpack	Accessories	3	1499	10	4047.3	UPI
Ishita Gupta	Pune	Yoga Mat	Sports	3	999	5	2847.15	UPI
Kavya Singh	Gurugram	Face Wash	Beauty	2	449	5	853.1	Debit Card
Aarav Gupta	Gurugram	Wireless Headphones	Electronics	3	2499	5	7122.15	Credit Card
Pooja Srivastava	Lucknow	Formal Shirt	Fashion	3	1199	15	3057.45	Credit Card
Yash Agarwal	Noida	Face Wash	Beauty	3	449	0	1347.0	Debit Card
Yash Singh	Bengaluru	Air Fryer	Home & Kitchen	2	5299	0	10598.0	Credit Card
Aditya Singh	Bengaluru	Dumbbells Set	Sports	1	1899	15	1614.15	Credit Card
Kavya Singh	Noida	Laptop Backpack	Accessories	2	1499	15	2548.3	Cash on Delivery
Meera Mishra	Bengaluru	Running Shoes	Fashion	3	2299	5	6552.15	Debit Card
Yash Verma	Bengaluru	USB-C Cable	Electronics	1	499	0	499.0	Debit Card
Rohan Singh	Gurugram	Cotton T-Shirt	Fashion	3	699	0	2097.0	UPI
Kavya Verma	Ahmedabad	Smart Watch	Electronics	1	3299	5	3134.05	Net Banking
Priya Verma	Chennai	Dumbbells Set	Sports	1	1899	15	1614.15	Net Banking
Kavya Singh	Noida	Dumbbells Set	Sports	2	1899	0	3798.0	UPI
Kabir Patel	Gurugram	Non-Stick Pan	Home & Kitchen	2	1399	10	2518.2	UPI
Pooja Srivastava	Lucknow	Water Bottle	Home & Kitchen	3	799	5	2277.15	Credit Card
Rohan Singh	Pune	Yoga Mat	Sports	1	999	10	899.1	Debit Card
Simran Khan	Lucknow	Yoga Mat	Sports	3	999	0	2997.0	Net Banking
Aditya Singh	Bengaluru	Laptop Backpack	Accessories	2	1499	10	2698.2	Credit Card
Kavya Verma	Ahmedabad	Water Bottle	Home & Kitchen	1	799	10	719.1	Cash on Delivery
Kabir Patel	Gurugram	Face Wash	Beauty	3	449	10	1212.3	Credit Card
Arjun Verma	Jaipur	Sunscreen SPF 50	Beauty	3	699	15	1782.45	Debit Card
Kavya Verma	Ahmedabad	Dumbbells Set	Sports	3	1899	15	4842.45	UPI
Aditya Singh	Bengaluru	Smart Watch	Electronics	3	3299	0	9897.0	Credit Card

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Ishita Singh	Noida	Sunscreen SPF 50	Beauty	1	699	15	594.15	UPI
Aditya Singh	Bengaluru	Sunscreen SPF 50	Beauty	3	699	15	1782.45	Debit Card
Aditya Sharma	Noida	Denim Jeans	Fashion	2	1599	0	3198.0	Debit Card
Priya Khan	Noida	Smart Watch	Electronics	1	3299	10	2969.1	Net Banking

3. Orders Dataset — continued

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Neha Sharma	Lucknow	Running Shoes	Fashion	3	2299	5	6552.15	Debit Card
Simran Khan	Lucknow	Formal Shirt	Fashion	1	1199	10	1079.1	Debit Card
Yash Verma	Bengaluru	Face Wash	Beauty	2	449	0	898.0	Debit Card
Pooja Srivastava	Lucknow	Power Bank	Electronics	2	1299	5	2468.1	Credit Card
Kunal Khan	Delhi	Denim Jeans	Fashion	3	1599	10	4317.3	UPI
Kabir Patel	Gurugram	Bluetooth Speaker	Electronics	1	1799	5	1709.05	Debit Card
Kunal Gupta	Kolkata	Non-Stick Pan	Home & Kitchen	3	1399	0	4197.0	UPI
Riya Srivastava	Hyderabad	Bluetooth Speaker	Electronics	2	1799	15	3058.3	Credit Card
Kavya Jain	Bengaluru	Formal Shirt	Fashion	2	1199	0	2398.0	Credit Card
Simran Khan	Lucknow	USB-C Cable	Electronics	2	499	15	848.3	Net Banking
Riya Srivastava	Hyderabad	Laptop Backpack	Accessories	1	1499	10	1349.1	Credit Card
Kunal Khan	Delhi	Cotton T-Shirt	Fashion	2	699	0	1398.0	UPI
Ishita Gupta	Pune	Dumbbells Set	Sports	1	1899	0	1899.0	Debit Card
Arjun Mishra	Pune	Wireless Headphones	Electronics	3	2499	0	7497.0	Debit Card
Neha Sharma	Lucknow	Air Fryer	Home & Kitchen	3	5299	15	13512.45	Net Banking
Yash Singh	Bengaluru	Denim Jeans	Fashion	1	1599	15	1359.15	UPI
Kavya Patel	Hyderabad	Air Fryer	Home & Kitchen	3	5299	5	15102.15	Net Banking
Kavya Singh	Gurugram	Sunscreen SPF 50	Beauty	1	699	5	664.05	Cash on Delivery
Yash Verma	Bengaluru	Running Shoes	Fashion	2	2299	10	4138.2	Debit Card
Aditya Sharma	Noida	USB-C Cable	Electronics	3	499	5	1422.15	Debit Card
Yash Verma	Ahmedabad	Water Bottle	Home & Kitchen	3	799	5	2277.15	Credit Card
Yash Verma	Ahmedabad	Water Bottle	Home & Kitchen	3	799	0	2397.0	Net Banking
Ishita Singh	Noida	Wireless Headphones	Electronics	2	2499	5	4748.1	Cash on Delivery
Kavya Patel	Hyderabad	Coffee Maker	Home & Kitchen	2	3499	10	6298.2	Credit Card
Kavya Singh	Noida	Smart Watch	Electronics	2	3299	15	5608.3	Net Banking
Kunal Khan	Delhi	Cotton T-Shirt	Fashion	3	699	5	1992.15	Credit Card

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Yash Singh	Bengaluru	Cotton T-Shirt	Fashion	2	699	15	1188.3	Cash on Delivery
Riya Gupta	Hyderabad	Water Bottle	Home & Kitchen	2	799	10	1438.2	Credit Card
Arjun Sharma	Gurugram	Non-Stick Pan	Home & Kitchen	1	1399	0	1399.0	Net Banking
Priya Khan	Noida	Face Wash	Beauty	1	449	0	449.0	Credit Card

3. Orders Dataset — continued

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Yash Verma	Bengaluru	Shampoo	Beauty	2	549	10	988.2	Net Banking
Sneha Jain	Noida	Laptop Backpack	Accessories	3	1499	10	4047.3	Debit Card
Priya Verma	Chennai	Dumbbells Set	Sports	3	1899	0	5697.0	Cash on Delivery
Ananya Srivastava	Pune	Denim Jeans	Fashion	2	1599	5	3038.1	UPI
Riya Srivastava	Hyderabad	Water Bottle	Home & Kitchen	3	799	0	2397.0	Credit Card
Neha Sharma	Lucknow	Non-Stick Pan	Home & Kitchen	2	1399	15	2378.3	Cash on Delivery
Arjun Verma	Jaipur	Water Bottle	Home & Kitchen	3	799	0	2397.0	Cash on Delivery
Arjun Sharma	Gurugram	Formal Shirt	Fashion	2	1199	10	2158.2	Net Banking
Arjun Mishra	Pune	Cotton T-Shirt	Fashion	2	699	10	1258.2	UPI
Kunal Khan	Delhi	Water Bottle	Home & Kitchen	3	799	0	2397.0	Credit Card
Meera Mishra	Bengaluru	Water Bottle	Home & Kitchen	3	799	15	2037.45	UPI
Aditya Singh	Bengaluru	Non-Stick Pan	Home & Kitchen	1	1399	10	1259.1	UPI
Ishita Gupta	Pune	Running Shoes	Fashion	2	2299	5	4368.1	Cash on Delivery
Yash Verma	Ahmedabad	Smart Watch	Electronics	2	3299	0	6598.0	Net Banking
Kavya Verma	Ahmedabad	Cotton T-Shirt	Fashion	3	699	0	2097.0	UPI
Yash Singh	Bengaluru	Running Shoes	Fashion	1	2299	15	1954.15	Credit Card
Kavya Singh	Noida	USB-C Cable	Electronics	3	499	0	1497.0	Debit Card
Aarav Gupta	Gurugram	Shampoo	Beauty	3	549	0	1647.0	Debit Card
Rohan Singh	Gurugram	Formal Shirt	Fashion	3	1199	10	3237.3	Credit Card
Rohan Singh	Gurugram	Cotton T-Shirt	Fashion	1	699	5	664.05	Net Banking
Kavya Singh	Gurugram	Bluetooth Speaker	Electronics	2	1799	15	3058.3	Debit Card
Vivaan Mishra	Jaipur	Coffee Maker	Home & Kitchen	1	3499	10	3149.1	UPI
Yash Agarwal	Noida	Yoga Mat	Sports	3	999	0	2997.0	Credit Card
Yash Verma	Bengaluru	Denim Jeans	Fashion	3	1599	15	4077.45	Cash on Delivery
Rohan Singh	Gurugram	Coffee Maker	Home & Kitchen	1	3499	0	3499.0	Cash on Delivery
Priya Khan	Noida	Shampoo	Beauty	3	549	10	1482.3	UPI

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Aditya Sharma	Noida	Dumbbells Set	Sports	1	1899	5	1804.05	Debit Card
Kavya Patel	Hyderabad	Denim Jeans	Fashion	3	1599	0	4797.0	Credit Card
Simran Khan	Lucknow	Dumbbells Set	Sports	3	1899	0	5697.0	Cash on Delivery
Aanya Sharma	Bengaluru	Wireless Headphones	Electronics	1	2499	5	2374.05	Cash on Delivery

3. Orders Dataset — continued								
customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Simran Khan	Lucknow	Sunscreen SPF 50	Beauty	2	699	10	1258.2	Net Banking
Kabir Patel	Gurugram	Coffee Maker	Home & Kitchen	2	3499	10	6298.2	Debit Card
Arjun Mishra	Pune	Running Shoes	Fashion	2	2299	0	4598.0	Net Banking
Neha Mishra	Hyderabad	Running Shoes	Fashion	3	2299	10	6207.3	Net Banking
Kavya Patel	Hyderabad	USB-C Cable	Electronics	1	499	5	474.05	Debit Card
Meera Singh	Gurugram	Face Wash	Beauty	1	449	5	426.55	UPI
Ananya Srivastava	Pune	Power Bank	Electronics	3	1299	10	3507.3	UPI
Ananya Srivastava	Pune	Dumbbells Set	Sports	2	1899	5	3608.1	UPI
Kavya Singh	Noida	Smart Watch	Electronics	2	3299	10	5938.2	Net Banking
Kavya Verma	Ahmedabad	Power Bank	Electronics	3	1299	5	3702.15	Credit Card
Sneha Jain	Noida	Shampoo	Beauty	3	549	15	1399.95	Net Banking
Kunal Gupta	Kolkata	Formal Shirt	Fashion	3	1199	10	3237.3	Net Banking
Kavya Verma	Ahmedabad	USB-C Cable	Electronics	1	499	0	499.0	UPI
Simran Khan	Lucknow	Face Wash	Beauty	1	449	0	449.0	Cash on Delivery
Kavya Patel	Hyderabad	Formal Shirt	Fashion	1	1199	0	1199.0	Debit Card
Priya Khan	Noida	Cotton T-Shirt	Fashion	2	699	15	1188.3	Credit Card
Kabir Patel	Gurugram	Face Wash	Beauty	1	449	0	449.0	Credit Card
Kavya Verma	Ahmedabad	Formal Shirt	Fashion	3	1199	15	3057.45	Cash on Delivery
Priya Khan	Noida	Formal Shirt	Fashion	3	1199	10	3237.3	Net Banking
Aanya Sharma	Bengaluru	Smart Watch	Electronics	3	3299	0	9897.0	Debit Card
Riya Patel	Delhi	Smart Watch	Electronics	1	3299	15	2804.15	UPI
Priya Verma	Chennai	Bluetooth Speaker	Electronics	1	1799	10	1619.1	Cash on Delivery
Neha Mishra	Hyderabad	Non-Stick Pan	Home & Kitchen	3	1399	10	3777.3	Cash on Delivery
Yash Verma	Ahmedabad	Coffee Maker	Home & Kitchen	3	3499	0	10497.0	Debit Card
Aanya Sharma	Bengaluru	Water Bottle	Home & Kitchen	3	799	0	2397.0	UPI
Riya Patel	Delhi	USB-C Cable	Electronics	2	499	15	848.3	Net Banking

customer_name	city	product_name	category	quantity	unit_price	discount_pct	total_amount	payment_method
Yash Verma	Ahmedabad	Running Shoes	Fashion	3	2299	5	6552.15	Cash on Delivery
Kunal Gupta	Kolkata	Running Shoes	Fashion	2	2299	15	3908.3	Debit Card
Aanya Sharma	Bengaluru	Wireless Headphones	Electronics	3	2499	15	6372.45	Credit Card
Sneha Jain	Noida	Face Wash	Beauty	2	449	0	898.0	UPI

4. SQL Project Questions

No.	Business Question
1	Find total revenue from all non-cancelled orders.
2	Find total number of orders and total quantity sold.
3	Calculate average order value.
4	Find the top 10 customers by total spending.
5	Find the top 10 best-selling products by quantity.
6	Find revenue generated by each product category.
7	Find monthly revenue for 2025.
8	Find the most frequently used payment method.
9	Calculate the percentage of cancelled orders.
10	Find customers who placed more than 3 orders.
11	Find products whose sales quantity is above the average product sales.
12	Calculate total discount given by category.
13	Find the city with the highest revenue.
14	Calculate average delivery time for delivered orders.
15	Rank products by revenue within each category using a window function.
16	Find each customer's first and most recent order date.
17	Find repeat customers who purchased in at least two different months.
18	Calculate running monthly revenue using a window function.
19	Find the top 3 customers in each city by spending.
20	Create a customer segment using CASE WHEN based on total spending.

Suggested database tables: customers, products, categories, orders, order_items, payments. For a normalized project, split the provided order-level CSV into these related tables using primary and foreign keys.